

MA1521 L1 Week 04 Wed 20260902

Find the point(s) on each of the following surfaces which is closest to the origin.

a) $xyz^2 = 1$

b) $x^2 - yz = 1$

A rectangular produce box is to be made of cardboard; the sides of single thickness, the ends of double thickness, and the bottom of triple thickness, with the top left open. Its volume is to be 1 cubic foot; what should be its proportions in order to use the least cardboard?

Find the extremal point of $x^2 + 2xy + 4y^2 + 6$ and show it is a minimum point

Consider all planes passing through the point $(2,1,1)$ and such that the intercepts on the three coordinate axes are all positive.

For which of these planes is the product of the three intercepts smallest?

(Hint: take the plane in the form $z = ax + by + c$, where a and b are the independent variables.)

For each of the following functions, find the critical points, and classify them using the 2nd-derivative criterion.

a) $x^2 - xy - 2y^2 - 3x - 3y + 1$ b) $3x^2 + xy + y^2 - x - 2y + 4$ c) $2x^4 + y^2 - xy + 1$

- a) Find the maximum and minimum points of the function $2x^2 - 2xy + y^2 - 2x$ on the rectangle $R : 0 \leq x \leq 2; -1 \leq y \leq 2$; using this information, determine the type of the critical point.
- b) Confirm the critical point type by using the second derivative test.

Find by the method of least squares the line which best fits the three data points

a) $(0, 0), (0, 2), (1, 3)$

b) $(1, 1), (2, 3), (3, 2)$

Use least squares to fit a second degree polynomial exactly through the points $(-1, -1), (0, 0), (1, 3)$